Appau Gideon Kofi Amo

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Education

- **University of Science and Technology of China (USTC)** Suzhou, China
M.S. in Software Engineering (AI/ML concentration); CGPA 3.13/4.3 *Sep. 2024 – Jul. 2027 (Expected)*
Thesis: *Class-Adaptive Focal Sharpness-Aware Minimization for Long-Tailed Visual Recognition.* Advisor: Assoc. Res. Pengkun Wang.

- **University of Education, Winneba** Ghana
B.Sc. in Information Technology Education; CGPA 3.61/4.0 *Aug. 2018 – Oct. 2022*
First Class Honours, Top 10%. Advisor: Dr. William Asiedu.

Experience

- **University of Science and Technology of China** Suzhou, China
Researcher; Advisor: Assoc. Res. Pengkun Wang *2025 – Present*
Proposed **CA-Focal-SAM**, a class-adaptive sharpness-aware minimization optimizer for long-tailed visual recognition that improves tail-class accuracy by 6.03% on CIFAR-10-LT (IR=200) with no architectural change or inference overhead. Designed **GammaAdaptor** (2 learnable scalars) mapping class frequency to per-class focal parameters, and a quadratic curriculum on the SAM perturbation radius that yields flatter minima verified by Hessian spectral analysis ($\lambda_{\max}$ 114.7 vs. Focal-SAM 121.4 via stochastic Lanczos quadrature). Further proposed **SFCA** (Sharpness-Feedback Classwise Allocation), combining a static log-scarcity prior with an online per-class SAM ascent-gap signal—smoothed by a robustly normalized exponential moving average and mapped through a bounded sigmoid—to allocate per-class sharpness during training with zero inference overhead; on CIFAR-100-LT (IR=200, Logit Adjustment) SFCA raises tail accuracy from 22.45% (Focal-SAM) to 24.49%, with class-identity and sign ablations (22.23%/22.89%) confirming the feedback is class-specific, and achieves the strongest corrupted tail accuracy on CIFAR-100-C among compared checkpoints. Evaluated with a multi-seed statistical protocol (mean ± std over 3 runs) on CIFAR-10/100-LT, ImageNet-LT, and CUB-200-LT; completed 20+ qualitative analyses including t-SNE embeddings, 2D/3D loss landscapes, and confusion matrices.

- **Future Interns** Remote
Machine Learning Intern *Mar. 2026 – Apr. 2026*
Built an end-to-end time-series forecasting system (6-month sales prediction) and an NLP ticket-classification pipeline using BERT/DistilBERT with sentiment analysis for priority routing; developed an AI recruitment tool combining spaCy/NLTK resume parsing with LLM-based job matching.

- **Akenten Appiah Menkah University** Kumasi, Ghana
IT Assistant *Nov. 2022 – Aug. 2023*
Automated data workflows in Python (Pandas, NumPy), cutting manual processing time by 40%; optimized SQL databases with 5,000+ records, improving query performance by 30%.

- **Amidarr A-mole** Ghana
Full-Stack Web Developer Intern *Aug. 2021 – Mar. 2022*
Engineered 3 full-stack web applications (Flask, JavaScript, MySQL), integrating the PayStack payment API into an e-commerce platform.

Publications

- Appau, G. K. A., & Wang, P. (2026). *Class-Adaptive Focal Sharpness-Aware Minimization for Long-Tailed Visual Recognition.* Manuscript in preparation (target: CVPR).
- Appau, G. K. A., & Wang, P. (2026). *Sharpness-Feedback Classwise Allocation for Long-Tailed Visual Recognition.* Manuscript in preparation (target: CVPR).

Selected Projects

- **RL Research Portfolio** (github.com/Amo-Gideon/rl-portfolio) *2026 – Present*
Reproduced Dueling/Double DQN from scratch and built an end-to-end RLHF pipeline (SFT, reward model, PPO alignment) for a 0.5B-parameter language model. On a PyBullet 2-link robot arm, PPO reaches 82%

single-target success while a learned world model (MLP ensemble) with CEM planning reaches 100% and stays robust under out-of-distribution physics shifts; obstacle clearance as a CEM cost term cut collision rates from 57% to 23% where five PPO training configurations failed.

- **Personalized Agentic RAG System** *USTC, 2025*

Adaptive retrieval system with autonomous agents, iterative self-correction, and user memory over domain-specific knowledge bases (LangGraph, LangChain, FAISS).

- **Campus Knowledge Base RAG Chatbot** *2025*

LLM-powered chatbot for semantic campus-information retrieval with a Chroma vector database and low-latency embeddings (Ollama, Gradio).

- **AI Study Coach** *2025*

Multi-agent learning assistant with 4 specialized AutoGen agents and Gemini 2.0 Flash for adaptive quiz generation.

- **Predictive House Price Modeling** *USTC, 2025*

Regression models achieving $R^2 = 0.89$ via hyperparameter tuning and outlier detection (scikit-learn).

- **CNN for FashionMNIST Classification** *USTC, 2025*

Custom PyTorch CNN reaching ∼90% accuracy through hyperparameter optimization.

Teaching and Mentorship

- **Peer Mentor, USTC Software Engineering** Suzhou, China

Mentor for junior master's students *2024 – 2025*

Guided PyTorch debugging and SAM optimizer implementations; explained gradient mechanics and loss-landscape visualization to labmates.

Awards

- **Chinese Government Scholarship (CSC), USTC** *2024*
- **First Class Honours, Top 10%, University of Education Winneba** *2022*
- **NVIDIA LLM RAG Agent Fundamentals** *2025*
- **DeepLearning.AI Advanced Learning Algorithms** *2025*
- **DeepLearning.AI Supervised Machine Learning: Regression & Classification** *2025*

Presentations

- **USTC Thesis Progress Seminar: CA-Focal-SAM for Long-Tailed Visual Recognition** *2026*
- **USTC Engineering Practice Defense: Homomorphic Encryption Deployment on FPGA** *2025*

Skills

Languages: Python, JavaScript, SQL, Java, Bash
ML/AI: PyTorch, scikit-learn, Transformers, LangChain, LangGraph, Ollama, AutoGen, FAISS, Chroma
LLM/NLP: BERT, DistilBERT, spaCy, NLTK, RAG agents
Data: Pandas, NumPy, Matplotlib, feature engineering, time-series forecasting
Web & Tools: Flask, REST APIs, Gradio, MySQL, Git, Linux, Docker, LaTeX

References

Available upon request.